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From Crisis to Control: Cambodia's Responses in Fighting Malaria

Malaria is an extremely prevalent disease that continues to have devastating effects on millions of people across the globe, especially in the tropical regions of Sub-Saharan Africa and parts of Asia. Based on the World Health Organization's statistics, at least as of 2022, there were about 608,000 deaths related to malaria and up to 249 million new cases of this disease (WHO, 2024). Among these affected regions is Cambodia, a country of about 17.7 million people situated in the heart of Southeast Asia. Emerging from decades of conflict and turmoil over the last century, from suffering under the French colonialization and enduring many civil wars to experiencing a very brutal genocide, Cambodia has made remarkable strides to become of the fastest developing nations. However, there are many sectors of Cambodian society that still need more improvement, particularly healthcare. The malaria endemic has posed a significant challenge to Cambodia's healthcare system. Fortunately, through a concerted effort by the government, international organizations and local communities, Cambodia has been able to survive through the worst of this crisis, although the disease still remains endemic in many parts of the country, especially in rural and remote areas where access to healthcare services is limited. Other factors, such as poverty, inadequate infrastructure and environmental conditions suitable for mosquitos, contribute to the persistence of malaria transmission in the country. Additionally, the emergence of drug-resistant strains of malaria has complicated the situation even more. Throughout this paper, I will examine Cambodia's overall healthcare system, as well as various interventions that have been implemented to tackle malaria, and evaluate the challenges and socioeconomic effects related to this issue.

The Cambodian healthcare system is somewhat similar to that of the US, which includes both public and private providers. According to the US's International Trade Administration, The public sector primarily focuses on health promotion, prevention and other essential care services, such as those related to reproductive, maternal and pediatric diseases, while private practitioners are more sought after for curative care (ITA, 2023). There are about 1,000 public and 8,000 private healthcare facilities (ITA, 2023). The healthcare expenditure has fluctuated over the years; it is reported that in terms of percentage of Cambodia's GDP, healthcare spending was at 6.4% in 2000 and increased up to 7.5% in 2011, but dropped back to 6.6% in 2019 (Hutt, 2019). However, in the following years, the COVID-19 pandemic happened, which forced the government to spend more than 10% of its GDP, or roughly $430 million on healthcare (ITA, 2023).

So far, Cambodia's healthcare has mostly improved from the past decades, but it is still mediocre compared to the rest of her neighbors. People who are wealthy enough would rather seek medical help in other countries, such as Thailand, Singapore, France and even the US since those countries have better overall healthcare (GIZ, 2021). Those who could not afford such high prices would go to Vietnam as healthcare over there is relatively cheap and of higher quality (GIZ, 2021). This goes to show that many Cambodian people are not having faith in their healthcare system. For poorer people, their only access to healthcare is their local hospital. Among the largest public hospitals are Calmette Hospital (for general care and emergency) and Kantha Bopha Hospital (for children), which are all located in the capital city of Phnom Penh. This can be a problem for people from rural or remote areas because they lack easy accessibility as they would have to travel for hours to reach these places. For a lot of times, these large hospitals become overcrowded with patients. For example, the Kantha Bopha Hospital is among the few that has adequate medical technologies and staffs, and so people all over the country would flock to this place, where hundreds, if not thousands, of whom would line up at the hospital front gate every morning to get an appointment with a doctor (Haffner, 2019).

Up until recently, malaria had been one of the deadliest diseases in Cambodian history. Thanks to its geographical location, the tropical climate has allowed for mosquitoes carrying the malaria disease to thrive. Specifically, they tend to live in forest regions, found in the western and easternmost parts of the country. Affected people—of whom, pregnant women and children are most vulnerable—tend to show symptoms like fever, chills, headache, vomiting and diarrhea, which could last for a week or up to a month; in worst cases, failure to seek treatment could cause kidney problems, seizures and even death (PRB, 2002). According to a journal article from 1992, it was estimated that “there [were] around half a million cases of malaria with 5,000–10,000 deaths per year” (Denis and Meek, 1992). It should be noted that in 1992, Cambodia barely got out of the wars and genocide and its healthcare system was in a state of neglect. As one would expect, people from the rural and remote areas were severely affected, as access to healthcare, especially during this time period, was little to none.

Aside from causing immediate health issues, this malaria endemic had put a lot of pressure on the economy. For one, people who were sick because of malaria would not be able to come to work, thereby reducing productivity and missing out on their much-needed income. Many children were affected and they could not attend school properly. One study shows that about 40% of instances where people lost their land had been attributed to diseases, although malaria is not explicitly specified (Ricci, 2012). Generally speaking, many poorer people could not afford to pay for malaria treatment, so they would resort to borrowing money or selling their properties, such as land or livestock (Ricci, 2012). Given these challenges, poor and rural people have been at most risk and in addition to providing quality healthcare, the government also needs to consider about accessibility and affordability for them to receive healthcare.

The late 1990s and early 2000s saw increased attention and resources being allocated to malaria control efforts, as the Cambodian government began implementing the National Malaria Control Program (known as the NMCP). At around the same time, Cambodia was introduced artemisinin-based combination therapies (ACT), which has been quite effective in treating malaria compared to the old antimalarial drugs (URC, 2020). However, Cambodia is also known to be the place where most drug-resistant forms of malaria evolve. Researchers found that “in the 1960s, resistance of P. falciparum to chloroquine emerged from [western Cambodia] before spreading to Asia and Africa and causing millions of deaths” and that beginning 2007, there have been reports of malaria being resistant to artemisinin and even the more recent piperaquine (Tripura, et al., 2017). Other common malaria treatments include mefloquine and artemether, but even they are becoming less and less effective particularly in these western provinces, again due to the uncontrollable, rapid mutations of the P. falciparum (PRB, 2002). Cambodia could not just rely solely on antimalarial drugs as it is shown that the parasites are evolving just as quickly as new medicine is invented.

The Cambodian healthcare system would not be able to handle this crisis without supervision and fundings from international NGOs, especially the WHO and UNICEF. In 1998, the WHO passed the Roll Back Malaria (RBM) policy, which granted a number of countries the necessary aid to combat malaria; Cambodia was among them and received a total of $30.4 million in loan, based on a 2002 report (Narasimhan and Attaran, 2003). In addition, University Research Co (URC) has made considerable effort into controlling malaria. They focus mainly in the western provinces of Pursat, Badombong and Pailun in which they divide into several operational districts and then conduct rigorous surveillance efforts to identify people who have malaria; they do this by “incorporating the 1-3-7 surveillance approach—to report confirmed malaria cases within one day, investigate confirmed cases within three days, and apply control measured to prevent further transmission within seven days, as well as directly observed therapy and follow-up microscopy” (URC, 2024). The group conducted this research on over 800,000 people and as a result, “98% of suspected malaria cases were tested… and 91% of confirmed positive cases received treatment” (URC, 2024). The whole project is huge and is still ongoing—the URC plans to help completely eliminate malaria in Cambodia by 2025. Another organization, Doctors Without Borders (or Médecins Sans Frontières), not only helps the local communities in a similar way to the URC, but also actively engages in awareness campaigns to bring attention about malaria. MSF teams would work closely with village elders to ensure that locals could understand the risks of getting malaria and the benefits of getting treatment (MSF, 2019).

The government also plays a huge part in spreading awareness for the population. As part of the school curriculum, students are required to learn about various diseases, such as malaria, dengue fever and HIV/AIDS. Furthermore, there are mass-advertisements on national television channels to encourage people to use vector controls, such as mosquito nets and insect repellents. One simple, yet ingenious, method is to breed small guppy fish in any water container, be it pots, jars, or even discarded tires, so that mosquitos would lay eggs there and consequently, the fish would eat up the larvae (TDR, 2019). These methods are easy and cheap, and prove to be quite successful in preventing mosquitos and helping to eliminate the sources of malaria. It is worth mentioning that Cambodia also suffers from dengue fever endemic in addition to malaria, and fortunately, these tactics are effective in curbing both diseases.

Another concern that needs to be addressed is that there could be information disparity when it comes to these awareness campaigns. There are dozens of different marginalized ethnic minorities in Cambodia, each with their own languages and dialects, such as Cham, Pnong and Kavet. Although Khmer is the main language in Cambodia, not everyone can read or write Khmer. As such, the National Center for Parasitology, Entomology and Malaria Control (CNM) made recorded messages in the eight most common ethnic languages to be put on loudspeakers at the back of motorcycles or cars as health workers would drive along their vehicles through villages in these remote areas (WHO, 2023). These ethnic minorities and those people living in the faraway countryside are at a huge disadvantage as access to proper healthcare is not as feasible for them. As malaria in Cambodia is known to be very adaptive to medicine and can transmit very easily, it is vital that the disease would not spread to these vulnerable communities, hence why the government incentivizes them to take a more proactive stance against it.

Thanks to these comprehensive preventative measures, the malaria crisis has mostly been averted. Thus far, there has been about 90% decrease in confirmed malaria cases between 2010 and 2020, from about 100,000 to a little less than 9,000 cases each year (Siv, et al., 2022). Another statistic shows that malaria cases went from 7.4 per 1,000 people in 2006 to just about 3.9 per 1,000 people in 2018 (Chhim, et al., 2021). In retrospect, this is a huge success, especially when one considers the number of malaria cases back in 1992, with half a million cases each year to now just several thousands. The number of deaths has also significantly been decreased. According to Statista, there were about 659 deaths related to malaria in 2010, but the number dropped to 345 deaths in 2017 (von Kameke, 2023). An official report from the CNM in 2023 puts the number of malaria-related death at zero (Mom, 2024). While these numbers offer a very optimistic outlook on Cambodia's future, it is important for the country to yet still remain vigilant against any strain of malaria that might evolve to overcome any latest medicine. So long as the government and these NGOs keep up the work that they have been currently doing, a complete elimination of malaria might one day be achieved.

Cambodia's decades-long fight against malaria should be considered a testament to how effective good strategies and collaborative efforts across all areas of society can be in preventing deadly diseases. The United States can certainly learn from Cambodia's experiences. Undoubtedly, the American healthcare offers comparatively much more higher quality services, thanks to having more advanced technology, sufficient funding and better staffing than Cambodia's. However, when looking at the social disparities in terms of healthcare accessibility and affordability, the US does not necessarily fare much better. The problem was very evident during the COVID-19 pandemic—there were significantly more deaths per 100,000 people for marginalized minorities, such as Native Americans, Hispanics, Pacific Islanders and African Americans, when adjusted for age distribution differences (Gawthrop, 2023). Another thing to point out is that up to over a million Americans died due to COVID-19, the highest number out of any other countries in the world (CDC, 2024). There are several factors contributing to this staggering number, such as lack of adequate care and proper management in the healthcare system, as well as preventative policies not being followed in certain states. One study finds out that “of the 8 states with at least 75% mask adherence, none reported a high COVID-19 rate [but] states with the lowest levels of mask adherence were most likely to have high COVID-19 rates” (Fischer, et al., 2021). Moreover, as of 2021, states with higher mask adherence had about 109.26 cases per 100,000 people, while those that do not had about 249.99 cases per 100,000 people (Fischer, et al., 2021). Although the COVID-19 pandemic in the US was effectively over, thanks to the vaccines, the casualties from this pandemic could have been reduced if the US government had implemented a more aggressive campaign to control the spread and bring awareness to the disease. Following the guidelines properly is also a big responsibility on the part of the public.

Overall, Cambodia’s strong commitment to fight off malaria that has plagued the country for decades has finally paid off. Despite numerous setbacks that the country has been facing, from insufficient funding for its healthcare system to contrasting disparities in terms of quality and access between rural and urban regions, its frontline workers have worked extremely hard to close the gaps and make sure anyone that is affected gets their much-needed treatment. Thanks to aggressive programs by the government and various international health organizations, as well as extensive community engagement, Cambodia manages to get the number of malaria cases to a very low number and the death toll as of recent to a zero, and is planning to completely eradicate any remnants of malaria in the next couple years. The fight is not over, yet, as there are many more deadly diseases still lingering in the country, like dengue fever and tuberculosis, which ultimately proves Cambodia needs to invest even more in its health infrastructure to meet international standards and give people better access to quality services.

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